

## Chapter 8

- p. 441 7LT Change "4.12" to "4.13".
- p. 446 19LB Replace "/" with "--".
- p. 446 17LB Change "port (IN:" to "port (Sig\_in:".
- p. 446 17LB Change two instances of "vector\_logic" to "logic\_vector".
- p. 446 Replace 10LB with the following text:

```
if Sig_in = k then Y(k) <= '1' else Y(k) <= '0'.
```

- p. 474 6LB Delete second semicolon at the end of the line.
- p. 474 7LB Replace "A: out" with "A: buffer".
- p. 475 19LB Change "else if" to "elsif".
- p. 475 16LB Replace "clock" with "Start".
- p. 475 9LB and 10LB Replace with the following text:

```
when S_1 => if (A2 = '1') then if (A3 = '1') then next_state <= S_2;  
            else next_state <= S_1; end if;
```

- p. 475 1LB, 2LB, 3LB Replace "0" with "'0'".
- p. 475 20LB Replace 20LB with the following text:

```
if(reset_b'event and reset_b = '0' then state <= S_idle;
```

- p. 475 19LB Replace "else if" with "elsif".
- p. 476 13LT Replace "A: out" with "A: buffer".
- p. 476 1LT, 2LT Replace "0" with "'0'".
- p. 476 4LT Replace " clr\_A\_F <= '1';" with "clr\_A\_F <= '1'; end if;"
- p. 476 5LT Delete the entire line.
- p. 476 6LT, 7LT Replace 6LT and 7LT with the following text:

```
when S_1      => incr_A <= '1'; if A2 = '1' then set_E <= '1';  
                else clr_E <= '1' end if;
```

- p. 476 24LT Replace "then <= A + 1; end if" with "then A <= A + "0001"; end if".
- p. 479 11LT Replace "Design Example" with "Design\_Example".
- p. 479 17LT, 19LT Replace "std logic" with "std\_logic"
- p. 479 17LT Delete period at end of line.
- p. 479 22LT Replace "sig" with "end component;"
- p. 479 4LB Replace "<= '1';" with "<= '1' after 5 ns;"
- p. 479 2LB, 1LB, p. 480 1LT, 2LT, 3LT Replace with the following text:

```
process begin  
  t_clock <= '0';  
  for k in 0 to 32 loop  
    wait for 5 ns;  
    count <= count + '1';  
    t_clock <= not t_clock;  
  end loop;
```

```
wait;  
end process;
```

p. 485 14LB, 15LB Replace 14LB, 15LB with the following text:

```
if clock'event and clock '1 then if clear = '1' then A <= 0;  
elsif incr = '1' then A <= A + "0001";
```

p. 485 4LB Replace "if RST = '0'" with "if RST'event **and** RST = '0'"

p. 486 1LB Replace "0" with "'0'".

p. 486 1LT Replace "(Q, Q\_not" with "(Q: **buffer** std\_logic; Q\_not"

p. 501 3LB Single to double quotes - Replace "'001'" with "'001'".

p. 501 2LB Single to double quotes - Replace "'010'" with "'010'".

p. 501 1LB Single to double quotes - Replace "'100'" with "'100'".

p. 502 15LB Delete "**begin**".

p. 507 5LB Replace 5LB with the following text:

```
end component;  
begin
```

p. 508 7LT Replace "'10111'" with "'10111'".

p. 508 8LT Replace "'10011'" with "'10011'".

p. 508 17LT Replace "0" with "'0'".

p. 512 3LT Replace "if Q(0)" with "if Q(0) = '1'".

p. 523 18LB Replace 18LB with the following text:

```
generic (R1_size: positive := 8; R2_size: positive := 4);
```

p. 523 11LB Add the following text at the end of the line: **end component;**

p. 523 9LB Add the following text to the end of the line: **end component;**

p. 524 1LT Change "controller to "Controller".

p. 524 2LT Replace "Ready," with "Ready **buffer**;"

p. 524 6LT, 7LT Change "**constant**" to "**signal**".

p. 525 3LT Replace 3LT with the following text:

```
M2: decoder_2x4_df port map (D => Decoder_out, A => G1, B => G0, enable => GND);
```

p. 525 4LT Replace "Ready => Sig1" with "Sig1 => Ready".

p. 525 4LT Replace "Start => Sig2" with "Sig2 => Start".

p. 525 15LT Replace 15LT with constant Zero: std\_logic := '0';

p. 525 16LT, 17LT Replace with the following text:

```
signal PWR: std_logic := '1';  
signal GND: std_logic := '0';
```

p. 525 11LB Replace "CLK, RST, Reset\_b" with "CLK, RST\_b".

p. 525 6LB Replace 6LB with the following text:

```
process (R1) begin
```

```

Zero <= '1';
for k in 0 to R1_size-1 loop if R1(k) = '1' then Zero <= '0'; end if; end loop;
end process;

```

p. 525 18LT Replace "Shift\_Reg"

with "Shift\_Reg **generic** (R1\_size : **positive**)"

p. 525 22LT Replace "Counter"

with "Counter **generic** (R2\_size: **positive**)"

p. 525 4LB Replace "**port map**" with "**generic map** (R1\_size => 8) **port map**"

p. 525 2LB Replace "**port map**" with "**generic map** (R2\_size => 4) **port map**"

p. 526 5LT Replace 5LT with "**generic** R1\_size: **positive**;"

p. 526 6LT Replace "R1: out" with "R1: **buffer**".

p. 526 12LT, 13LT Replace 12LT, 13LT with the following text:

```

signal PWR: Std_Logic := '1'

```

```

signal GND: Std_Logic := '0'

```

p. 526 17LT Replace 17LT with

```

if reset_b'event and reset_b = '0' then R1 <= '0';

```

p. 526 20LT Replace "SIO" with "SI\_O".

p. 526 7LB Replace 7LB with the following text:

```

if reset_b'event and reset_b = '0' then R2 <= '0';

```

p. 526 15LB Replace 15LB with the following text:

```

generic (R2_size: positive);

```

p. 526 11LB Replace "Counter" with "Counter **is**".

p. 526 4LB Replace "R2 + 1;" with "R2 + '1';".

p. 527 2LT Replace 2LT with the following text:

```

port (Q: out Std_Logic; D, CLK, RST_b: in Std_Logic);

```

p. 527 6LT Replace "**process**" with "**process** (CLK, RST\_b) **begin**".

p. 527 7LT Replace 7LT with "**if** RST\_b'event **and** RST\_b = '0' **then** Q <= '0';"

p. 527 8LT T In two instances, replace "clock" with "CLK".

p. 527 13LT Replace 13LT with the following text:

```

port (Q: buffer Std_Logic; Q_b: out Std_Logic; D, CLK, RST_b: in Std_Logic);

```

p. 527 17LT Replace "**process begin**" with "**process** (CLK, RST\_b) **begin**".

p. 527 18LT Replace 18LT with the following text:

```

if RST_b'event and RST_b = '0' then Q <= '0';

```

p. 527 13LB, 14LB Delete 13LB and 14LB.

p. 527 6LB Replace "**port**" with "**generic map** (R1\_size => 8; R2\_size => 4) **port**".

p. 527 13LB Replace "0);" with "0));".

p. 527 14LB Replace "R1:" with "(R1:".

p. 527 14LB Replace "0);" with "0));".

p. 528 17LT Replace ""1111111" with ""11111111".

p. 528 25LT Replace "/" with "--".

p. 535 26LT Replace "precluded" with "precluded in SystemVerilog".

p. 543 No italics - In Problem 8.15, replace "*Simulate*" with "Simulate".

- p. 547 8LT Replace two instances of "RA == 0" with "RA = '0'".
- p. 547 8LT Replace "E := 1" with "E := '1'".
- p. 547 8LT Replace "E <= 1" with "E <= '1'".
- p. 547 9LT Replace "E := 0" with "E := '0'". Replace "E <= 0" with "E <= '0'".
  
- p. 547 14LT In 8.32 change "following always block" to "following **always** block".
- p. 551 12LT In 8.44, replace "in3" with "in\_3".
- p. 551 In 8.45, replace "(Verilog)" with "SystemVerilog".
- p. 551 In 8.46, replace "(Verilog)" with "SystemVerilog".
- p.551 1LB Delete 1LB
- p. 551 In 8.44 replace **input** in\_1, in\_2, in3,"  
with **input** [7:0] in\_1, in\_2, in\_3, **input** clk,".
- p. 552 1LT, 2LT, 3LT Delete 1LT, 2LT, 3LT.
- p. 552 In 8.47 replace "code" with "SystemVerilog code".
- p. 552 In 8.48, replace "model" with "SystemVerilog model".
- p. 553 In 8.51 replace "s2, s, s4" with "s2, s3, s4".
- p. 643 15LT Single to double quotes - Replace "'10\_0011'" with "\"100011\"".
- p. 643 16LT Replace "'01\_0101'" with "\"010101\"".
- p. 643 2LB Replace "'0011'" with "\"0011\"".
- p. 643 17LT Replace "'01\_1010'" with "\"01\_1010\"".
- p. 644 1LB to 9LB Replace all single quotes with double quotes.
- p. 645 8LB Replace "'0000'" with "\"0000\"".
- p. 645 7LB Replace "D0 = 1" with "D0 = '1'".
- p. 645 6LB Replace "D0 = 0 and D1 = 1" with "D0 = '0' **and** D1 = '1'".
- p. 645 5LB Replace "D0 = 0 and D1 = 0 and D2 = 1"  
" with "D0 = '0' **and** D1 = '0' **and** D2 = '1'".
- p. 645 4LB Replace "'0001'" with "\"0001\"".
- p. 649 12LB Replace 12LB with the following text:  
**if** CLK'event **and** CLK = '1' **then** Q <= S1 & Q(3:1); **end if**;
- p. 649 6LB Add the following text to the end of 6LB: **end component**;
- p. 651 In 9LB through 4LB replace "then" with **then**".
- p. 651 8LT Replace "Clock = 1" with "Clock = '1'".
- p.651 9LT Replace 9LT with **"if** Load = '1' **then** A <= "0001"".
- p. 651 16LB Replace "std\_logic\_vector (3 **downto** 0)" with **"integer range** (3 **downto** 0".
- p. 651 11LB Replace "reset = 1" with "reset = '1'".
- p. 651 4LB to 9LB Delete quotes.p. 651 4LB to 7LB Change "==" to "="
- p. 653 Replace 12LT to 21LT with the following text:

```

constant s0: Std_Logic_Vector (3 downto 0) = "0000";
constant s1: Std_Logic_Vector (3 downto 0) = "0001";
constant s2: Std_Logic_Vector (3 downto 0) = "0010";
constant s3: Std_Logic_Vector (3 downto 0) = "0011";
constant s4: Std_Logic_Vector (3 downto 0) = "0100";
constant s5: Std_Logic_Vector (3 downto 0) = "0101";
constant s6: Std_Logic_Vector (3 downto 0) = "0110";

```

**constant** s7: Std\_Logic\_Vector (3 **downto** 0) = "0111";  
**constant** s8: Std\_Logic\_Vector (3 **downto** 0) = "1000";  
 p. 653 19LB Replace "clock = 1" with "clock = '1'".  
 p. 653 15LB Replace "1'b0;" with "'1'";  
 p. 653 13LB Replace "if (start)" with "if start = '1'".  
  
 p. 654 1LT and 2LT: Delete 1LT and 2LT.  
 p. 658 4LB, 3LB Replace "= 1" with "= '1'".  
 p. 659 20LT Bold: **when** S1  
 p. 659 10LB Replace "in std\_logic\_vector" with "**in** std\_logic\_vector".  
 p. 659 9LB Replace "in" with "**in**".  
 p. 658 8LB Replace "out" with "**out**".  
 p. 659 15LT, 16LT, 17LT Replace "<= 0;" with "<= '0'";  
 p. 659 19LT Replace "start = 1" with "start = '1'".  
 9. 659 19LT Replace "<= 1;" with "<= '1'";  
 p. 659 21LT Replace two instances of "= 1" with "= '1'".  
 p. 664 18LB Replace 18LB with  
     **constant**    s0: Std\_Logic := '0';  
     **constant**    s1: Std\_Logic := '1';  
 664 15LB Replace "= 0" with "= '0'".  
 p. 664 14LB Replace "= 1" with "= '1'".  
 p. 664 8LB, 9LB, 10LB, 11LB Replace "<= 0" with "<= '0'".  
 p. 664 6LB Replace two instances of "= 1" with "= '1'".  
 p. 664 5LB, 4LB Replace "<= 1" with "<= '1'".  
 p. 664 3LB Replace "<= 0" with "<= '0'".  
 p. 665 10LT Insert "**end component**" at the end of 10LT.  
 p. 665 13LT Insert "**end component**:" at the end of 13LT.  
 p. 668 4LB Change "8\_8" to "8\_9".  
 p. 669 14LB to 10LB Replace "<= 0" with "<= '0'".  
 p. 669 17LB Replace "reset\_b = 0" with "reset\_b = '0'".  
 p. 669 16LB Replace "clock = 1" with "clock = '1'".  
 p. 669 9LB Replace "**case** state" with "**case** state **is**".  
 p. 670 1LT Replace "<= 1" with "<= '1'".  
 p. 670 2LT Replace "<= 0" with "<= '0'".  
 p. 674 2LT Replace "= 0" with "= '0'".  
 p. 674 3LT Replace "= 1" with "= '1'".  
 p. 674 6LT, 7LT Replace "<= 0" with "<= '0'".  
 p. 674 9LT Replace two instances of "= 1" each with "= '1'".  
 p. 674 22LT After 22LT insert new line: **end component**;  
 p. 674 24LT After 24 LT insert new line: **end component**;